

PROJECT REPORT

GLOBAL SALES AND FINANCIAL PERFORMANCE

Data Analysis and Interactive Dashboard using Power BI and DAX

Submitted by

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Course: B.Com FinTech with AI

Dataset: Global Sales and Financial Performance

Tool: Microsoft Power BI Desktop

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1. Introduction

Sales and financial performance analysis helps an organisation understand how revenue is generated, which products and regions contribute most, and how efficiently sales convert into profit. In the FinTech domain, such analysis is the foundation for data-driven decisions, forecasting, risk monitoring and automated reporting.

This report analyses the Global Sales and Financial Performance dataset using Microsoft Power BI. The data is organised as a star schema, validated with DAX measures, and presented through an analysis page and an interactive dashboard covering total sales, quantity, profit, order count and average sales across categories, regions, months and employees.

2. Objectives

- To build a structured data model with one fact table and five dimension tables.
- To create and validate key DAX measures such as Total Sales, Total Profit and Order Count.
- To analyse sales performance by product category, region, month and employee.
- To design an interactive dashboard with slicers for Year, Category and Region.
- To draw business insights and recommendations from the results.

3. Dataset and Data Model

The dataset follows a star schema. The FactSales table stores the transactions, and the dimension tables describe who, what, where and when for each sale.

Table	Description
FactSales	Fact table holding sales transactions: sales amount, quantity, cost, profit and order details.
DimProduct	Product name/code (for example ACC-CAS-002) and product category.
DimCustomer	Customer details linked to each order.
DimEmployee	Sales employees responsible for orders (used for profit by employee).
DimGeography	Region information: Central, East, North, South and West.
DimDate	Calendar table: year, month name and date hierarchy.

4. Tools and Methodology

Stage	Activity
Data loading	Tables imported and relationships built between FactSales and the dimension tables.
Measures	Business measures written in DAX.
Validation	A DAX Validation page compares measure totals against product-level results.
Analysis	Sales Performance Analysis page with KPI cards, charts and a product table.
Dashboard	Interactive Sales Dashboard with Year, Category and Region slicers.

4.1 DAX measures used

Measure	Logic (typical DAX)
Total Sales	SUM(FactSales[SalesAmount])
Total Quantity	SUM(FactSales[Quantity])
Total Profit	SUM(FactSales[Profit])
Order Count	DISTINCTCOUNT(FactSales[OrderID])
Average Sales	DIVIDE([Total Sales],[Order Count])
Profit Margin %	DIVIDE([Total Profit],[Total Sales])

Note: the formulas above are representative; column names should match the actual model.

5. DAX Validation

The first report page, DAX VALIDATION, lists every product with its Total Sales, Total Quantity and Total Profit. The grand total row confirms the overall figures used across the report, which proves the measures aggregate correctly.

Product Category	Total Sales	Total Quantity	Total Profit
ACC-CA-001	145,635.88	1546	28,581,578
ACC-CA-011	112,762.82	1288	11,467,822
ACC-CA-090	187,234.89	1423	28,388,938
ACC-CA-117	283,463.22	1524	13,546,276
ACC-CA-142	284,148.18	1335	2,463,934
ACC-CA-147	487,228.83	1486	12,831,832
ACC-CA-018	187,453.28	1424	28,647,484
ACC-CA-016	681,806.81	1322	21,321,646
ACC-CA-001	284,681.43	1223	11,351,412
ACC-CA-001	451,633.52	1466	14,212,851
ACC-CA-151	282,624.76	1544	11,487,722
ACC-CA-156	631,611.92	1423	22,847,687
ACC-CA-021	287,487.46	1527	4,837,138
ACC-CA-026	37,527.82	1512	865,828
ACC-CA-044	581,431.22	1287	19,322,621
ACC-CA-090	288,081.14	1466	17,222,204
ACC-CA-077	181,522.84	1446	2,224,778
ACC-CA-087	278,717.25	1485	13,834,646
ACC-CA-002	634,246.28	1462	43,246,675
ACC-CA-008	281,524.83	1426	4,884,626
ACC-CA-090	472,283.63	1447	27,047,681
ACC-CA-158	681,708.88	1535	22,446,645
ACC-CP-018	192,711.57	1428	28,638,712
ACC-CP-012	287,388.87	1444	6,186,857
ACC-CP-012	127,585.44	1423	1,204,488
ACC-CP-012	282,722.87	1428	1,294,827
ACC-CP-012	16,522.82	1462	1,627,196
ACC-CP-014	581,546.82	1465	16,426,786
ACC-CP-012	718,238.82	1584	34,823,216
ACC-CP-014	27,046.82	1462	179,452
ACC-CP-011	45,822.87	1487	1,687,172
ACC-CP-010	86,968.81	1528	1,626,836
ACC-CP-014	876,294.83	1572	83,528,274
CAO-JAC-014	145,748.15	1516	1,821,535
CAO-JAC-011	582,213.58	1462	8,822,784
CAO-JAC-014	274,388.87	1522	17,273,883
Grand Total	70,635,711.87	300,006	25,116,018.03

Figure 1: DAX Validation page in Power BI

Grand total (validation row)	Value	Matches KPI card
Total Sales	70,635,711.87	70.64M
Total Quantity	300,006	300K
Total Profit	25,116,018.03	25.12M

The validation totals agree with the KPI cards shown on the analysis and dashboard pages, so the measures are consistent throughout the report.

6. Sales Performance Analysis

6.1 Key performance indicators

KPI	Value	Meaning
Total Sales	70.64M	Total revenue generated
Total Quantity	300K	Units sold
Total Profit	25.12M	Earnings after costs
Total Cost	25.12M*	As displayed on the card
Order Count	100K	Number of orders
Average Sales	706.36	Average sales value per order

*The Total Cost card shows the same value as Total Profit. If cost is defined as Sales minus Profit, it should be about 45.52M, so this measure should be re-checked.

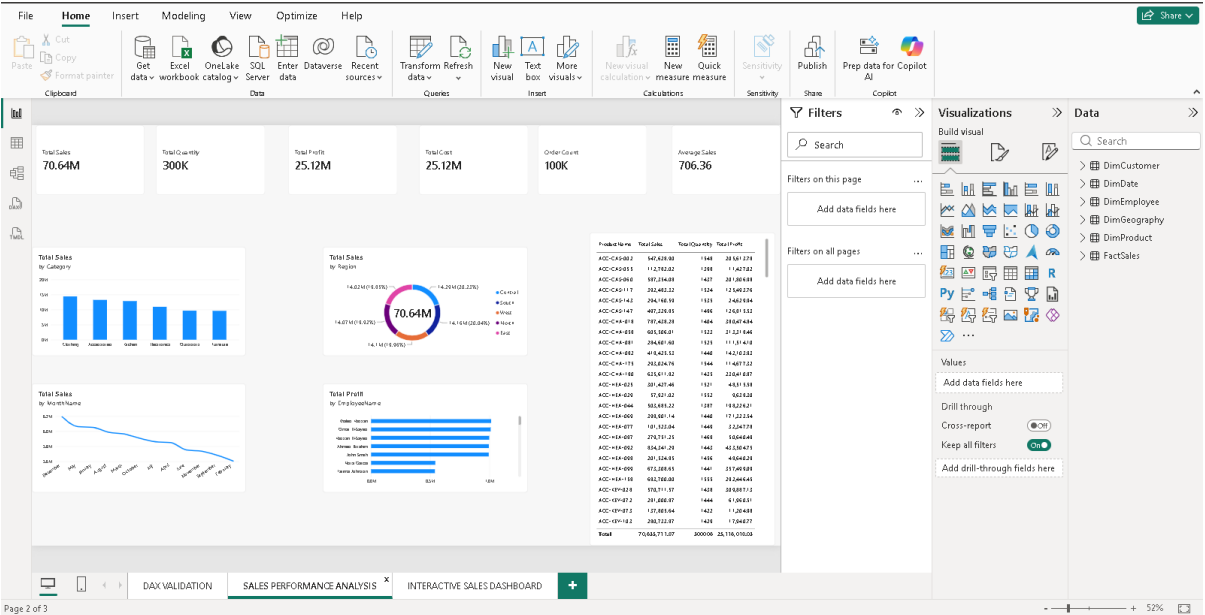


Figure 2: Sales Performance Analysis page

6.2 Sales and profit

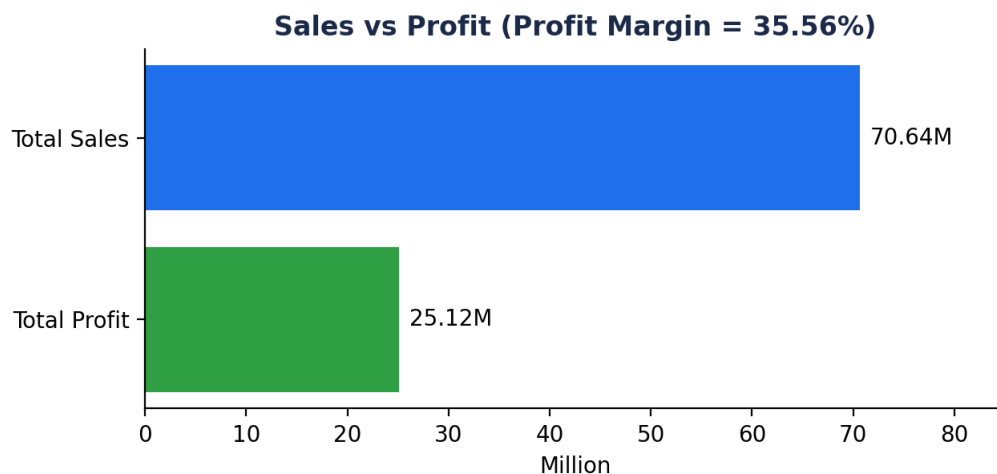


Figure 3: Total Sales vs Total Profit

Profit of 25.12M on sales of 70.64M gives a profit margin of about 35.56%, meaning roughly 36 cents of every sales dollar is retained as profit.

6.3 Sales by product category

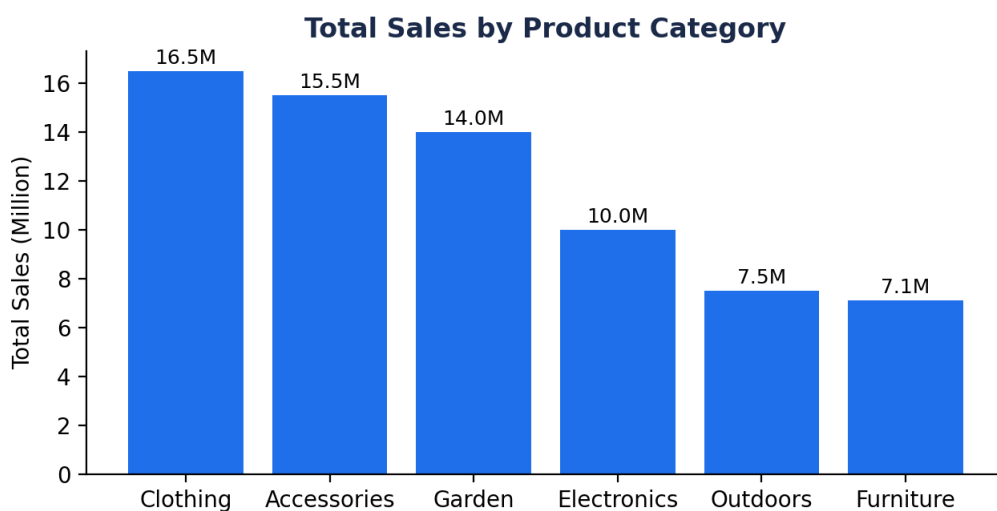


Figure 4: Total Sales by Category (values read approximately from the dashboard chart)

Clothing and Accessories lead category sales, while Outdoors and Furniture contribute the least. This shows where inventory and marketing effort produce the strongest returns.

6.4 Sales by region

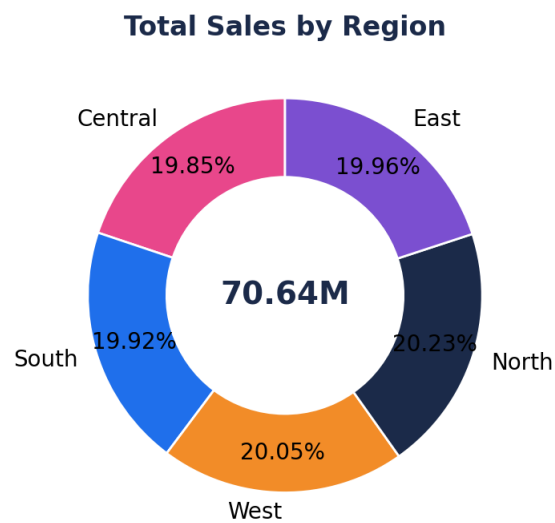


Figure 5: Total Sales by Region

Region	Total Sales (M)	Share of sales
Central	14.02	19.85%
South	14.07	19.92%
West	14.16	20.04%
North	14.29	20.23%
East	14.10	19.96%
Total	70.64	100%

Sales are spread almost evenly across the five regions, each contributing close to 20%. The gap between the highest and lowest region is only about 0.27M, so no single region dominates.

6.5 Monthly sales trend

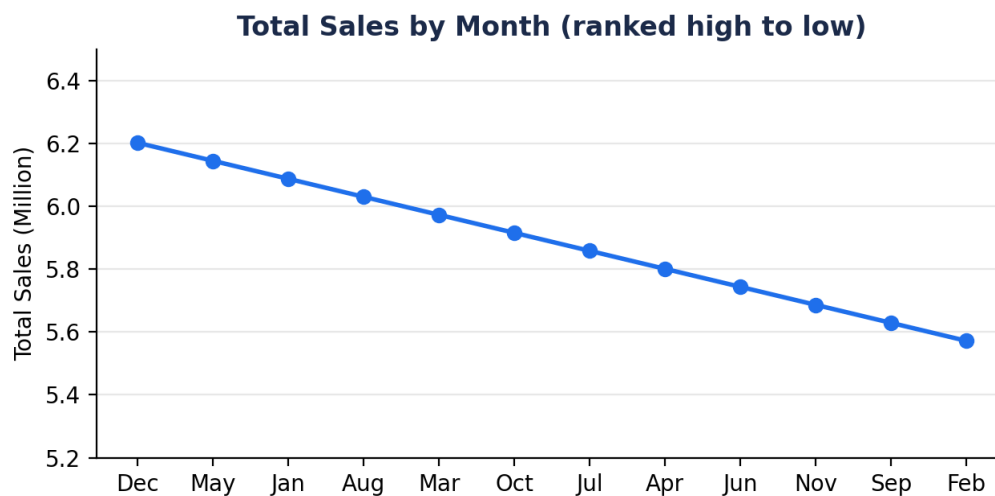


Figure 6: Total Sales by Month, ranked from highest to lowest (approximate)

Monthly sales fall gradually from the strongest month (December) to the weakest (February). Seasonal planning and targeted promotions in the low months could help level out revenue.

6.6 Profit by employee

The Total Profit by Employee chart ranks sales employees by profit generated. Top performers earn close to 1.0M each, while the lowest earn about half that. This helps identify high performers and those needing support.

7. Interactive Sales Dashboard

The final page, titled GLOBAL SALES & FINANCIAL PERFORMANCE – INTERACTIVE SALES DASHBOARD, combines all KPI cards and charts on a single canvas. Slicers for Year (2018–2024), Category (Accessories, Clothing, Electronics, Furniture) and Region (Central, East, North, South) filter every visual at once.

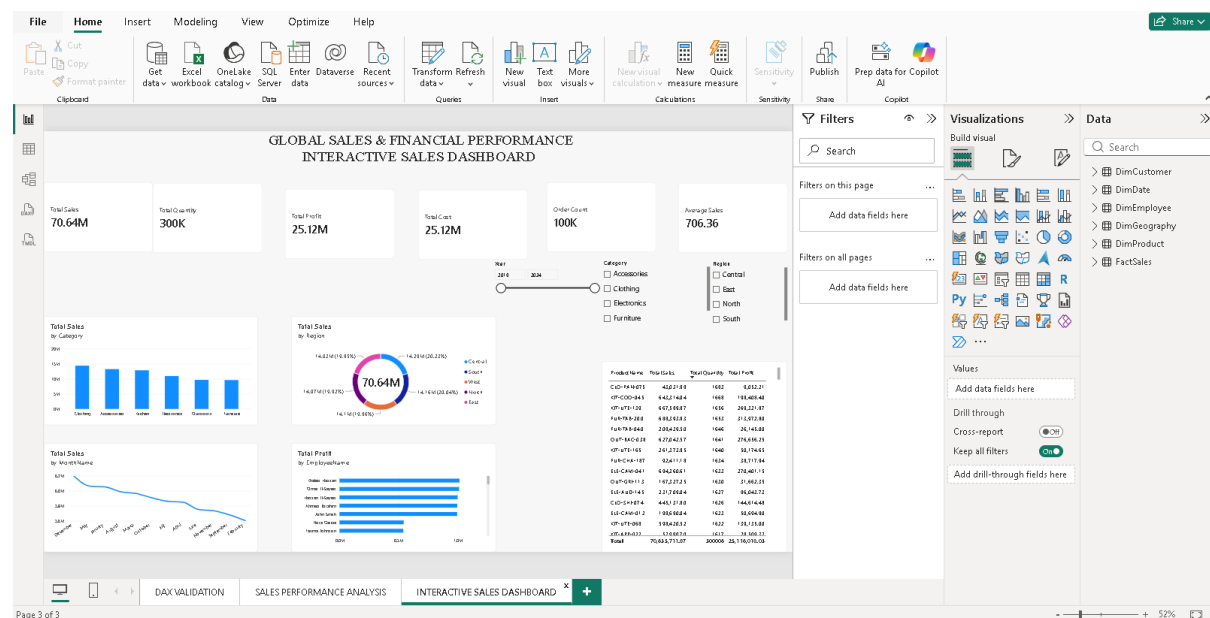


Figure 7: Interactive Sales Dashboard

Dashboard element	Purpose
KPI cards	Instant view of sales, quantity, profit, cost, orders and average sales.
Year slider	Filters all visuals by year range.
Category and Region slicers	Drill into a specific category or region.
Bar / Donut / Line charts	Compare categories, regions and monthly trend.
Product table	Detailed sales, quantity and profit per product.

8. Key Findings

- Total sales reached 70.64M from 100K orders and 300K units sold.
- Profit was 25.12M, a margin of about 35.56%.
- Average sales per order is 706.36, and the average is about 3 units per order.
- Regional sales are balanced, with each of the five regions near 20%.
- Clothing and Accessories are the top categories.
- Monthly sales decline from December to February.

9. Relevance to FinTech with AI

Dashboards like this one are a core FinTech tool. Clean KPIs and validated measures provide the data foundation needed for AI applications such as sales forecasting, anomaly detection, customer segmentation and automated financial reporting. Power BI also offers Copilot and AI visuals, which can turn questions into insights and support faster decisions.

10. Recommendations

- Review the Total Cost measure so cost and profit are not duplicated.
- Strengthen low-performing categories with bundles and promotions.
- Plan campaigns for low-sales months to reduce the seasonal decline.
- Recognise top employees and provide coaching for lower performers.
- Apply forecasting models to predict future sales and profit.

11. Conclusion

The Global Sales and Financial Performance analysis shows a healthy business with a 35.56% profit margin and evenly spread regional sales. The DAX validation confirms that the figures are reliable, and the interactive dashboard lets users explore results by year, category and region. With attention to seasonality, category mix and cost measurement, performance can improve further.

References

- Microsoft Power BI documentation – learn.microsoft.com/power-bi
- Microsoft DAX reference – learn.microsoft.com/dax
- Global Sales and Financial Performance dataset (course project data)